

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge



Source: arxiv.org/abs/2504.03553

Problem Statement

Current agent training blindly injects knowledge — like "flood irrigation" — regardless of situational need

1. Trajectory Imitation

Examples: ReAct, basic agent methods

Indiscriminately copies gold trajectories.

No self-awareness of when action is correct vs. needs reflection.

✗ **Pattern collapse on unexpected inputs**

2. Trial-and-Error Refinement

Examples: Reflexion

Relies on external feedback to refine.

Does not learn when to self-correct vs. when to seek knowledge.

✗ **Fragile planning, high inference cost**

3. Knowledge-Augmented

Examples: KnowAgent, WKM

Blindly injects 100% external knowledge.

No selectivity — wastes resources on steps that don't need it.

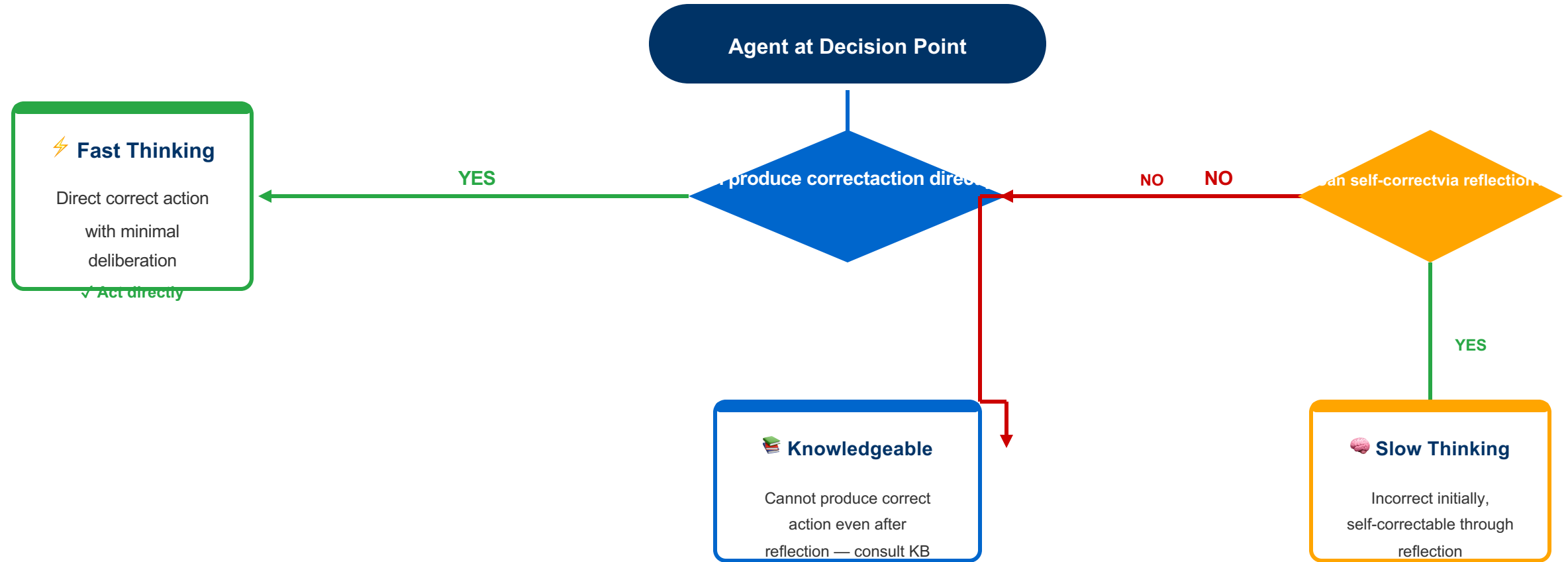
✗ **Indiscriminate = inefficient & suboptimal**

Humans possess situational self-awareness — the metacognitive ability to assess:

Can I act directly? → Need to reflect? → Must seek external knowledge?

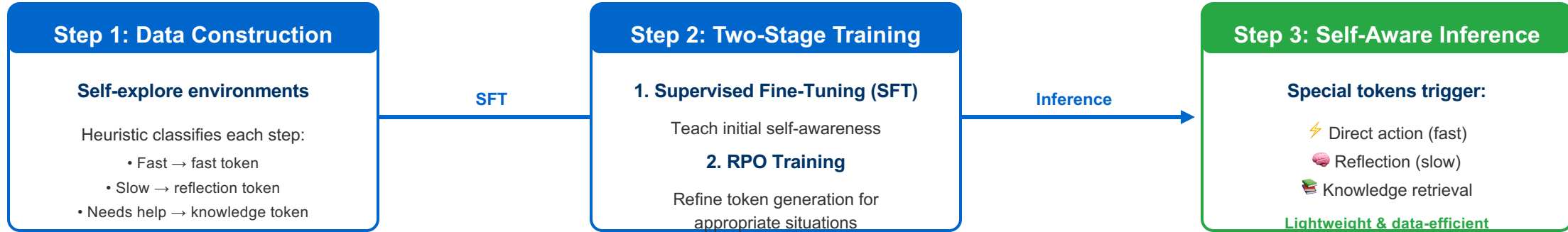
Three Situational Thinking Modes

KnowSelf defines three cognitive situations based on the agent's ability at each decision step



KnowSelf Framework

Three-stage pipeline: Heuristic labeling → Two-stage training → Self-aware inference



Key Innovation:

Special tokens mark situations — no expensive human annotation or massive external feedback required

Heuristic labeling on self-explored trajectories enables autonomous knowledge regulation

Qualitative Examples

KnowSelf correctly handles tasks where general reasoning models fail

Task: "Put a hot mug in cabinet"

✗ OpenAI-O1 (General Reasoning)

Incorrectly tries to take an egg from the microwave after opening it.

Fails: Object confusion in simple task

✓ KnowSelf (Llama-8B)

Uses Reflection token to realize the mug should be heated using the open microwave before placing in cabinet.

✓ Success: Proper action sequence

KnowSelf recognizes when reflection is needed

Task: "Put a saltshaker in drawer"

✗ DeepSeek-R1 (General Reasoning)

Reflects that the drawer is open and tries to close it first — which fails.

Fails: Over-complication of simple task

✓ KnowSelf (Llama-8B)

Applies Knowledge token — notes that the agent should place the object without removing unrelated items.

✓ Success: Correct knowledge application

KnowSelf recognizes when to seek knowledge

Key Insight: Situational self-awareness outperforms general reasoning — KnowSelf knows when to think, when to reflect, and when to seek knowledge.

ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge

KnowSelf uses far less knowledge (15% and 26%) yet outperforms methods using 100% knowledge

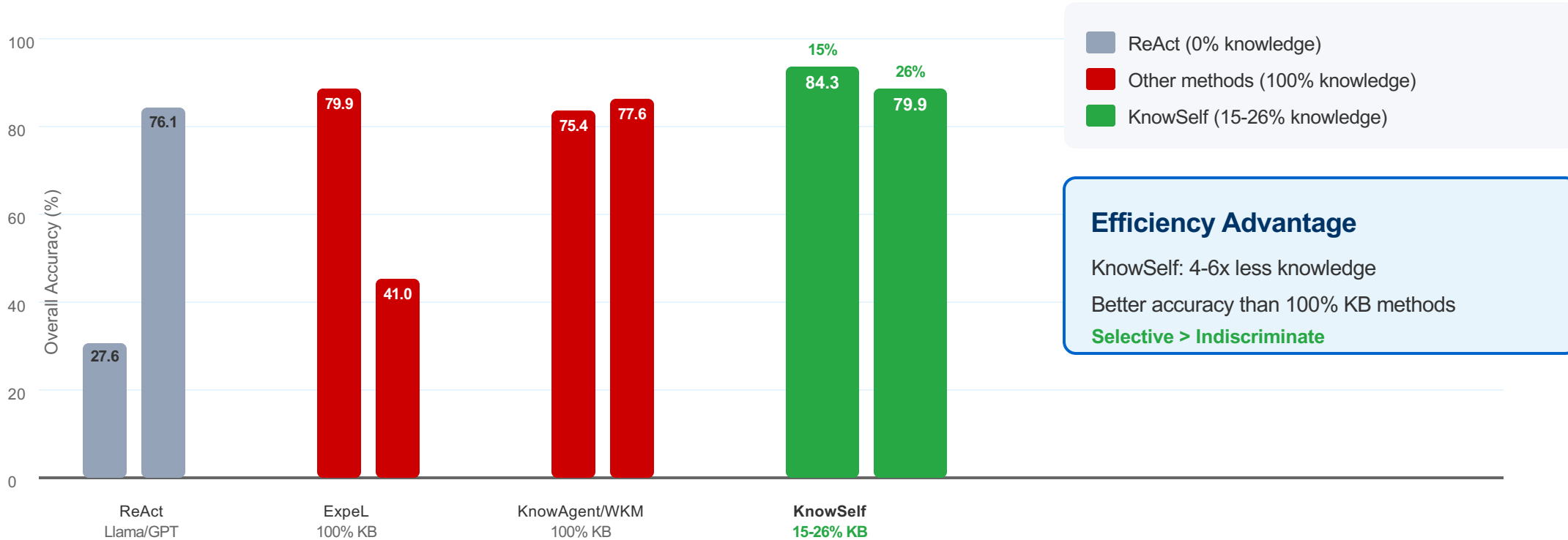
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	KnowSelf	15%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	KnowSelf	26%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Key Findings:

- Llama-8B KnowSelf (84.33%) surpasses GPT-4o ExpeL
- Gemma-2B KnowSelf (79.85%) matches GPT-4o ExpeL
- Only 15-26% knowledge vs 100% for competitors

Minimal Knowledge, Maximum Impact

KnowSelf uses only ~15-26% knowledge queries vs 100% for competing methods — yet achieves best scores



Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful

Methods using 100% knowledge often underperform KnowSelf, confirming that blanket application hurts rather than helps.

2

Agents can learn situational self-awareness via tokens

Lightweight heuristic labeling on self-explored trajectories enables autonomous cognitive resource regulation.

3

Llama-8B KnowSelf outperforms GPT-4o baselines

84.33% vs 76.12% (GPT-4o ReAct) and 79.85% (GPT-4o ExpeL) — selective use beats blanket injection.

4

Only 15-26% knowledge usage needed

vs 100% for competing methods — 4-6x efficiency gain without sacrificing accuracy.

5

Paradigm generalizes across model scales

Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines — demonstrating the approach scales down effectively while maintaining efficiency.

The three-situation taxonomy (fast/slow/knowledgeable thinking)

provides a principled framework for agents to autonomously regulate cognitive resources — analogous to human metacognition.